

EXPO MTPM Replacement — Technology Stack Decision Record

AISAVING-130 | Status: Pending Richard Patch Approval
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Purpose

This document records the evaluation of candidate technology stacks for the EXPO MTPM replacement application, documents the rationale for the recommended approach, and serves as the AC1 artifact for the *Tech Stack Selection & Dev Environment* story. Per NFR-5 and the requirements business rules, the tech stack must be documented and approved by the product owner before build begins.

Evaluation Criteria

Candidates were evaluated against the following requirements from the v3 requirements document:

Criterion	Source
Hosted within GDMS M365 GovCloud — no data leaves boundary	NFR-3
Authentication via Azure AD — no separate login	NFR-3, FR 6.1
Power BI external connectivity supported	FR 5.10, Power BI Connectivity section
Responsive under up to 250 concurrent users	NFR-1
Uses GDMS-approved, supported technologies	NFR-5
Maintainable by a small dev team with no dedicated DBA/DevOps	NFR-5
Source in version-controlled repository	NFR-5
Usable in GDMS-approved browsers without plugins	NFR-4

Candidate A — Recommended: .NET Core + Azure SQL Database + React

Overview

Layer	Technology
Frontend	React (TypeScript) — served as a static web app or from Azure App Service
API	ASP.NET Core Web API (.NET 8)
Database	Azure SQL Database (PaaS managed service)
Auth	Microsoft Identity / MSAL — Azure AD (GDMS M365 GovCloud)

Layer	Technology
Hosting	Azure App Service (API + frontend) on the GDMS GovCloud tenant
Source Control	Azure DevOps (GDMS GovCloud)

Evaluation Against Criteria

Criterion	Assessment
GovCloud boundary	Azure SQL Database, App Service, and Azure AD are all available in Azure Government Cloud. No data leaves the GDMS M365 GovCloud tenant.
Azure AD authentication	Native support. MSAL + Microsoft Identity Platform integrates directly with GDMS Azure AD. No separate login required.
Power BI connectivity	Azure SQL Database is natively supported as a Power BI data source. The existing Power BI dashboard can be reconnected to the new Azure SQL endpoint without modifying report logic. Supports both Direct Query and Import modes.
Concurrent user load (250)	Azure SQL Database scales independently of the application tier. Azure App Service scales horizontally. Well within capacity for this workload.
GDMS-approved technologies	.NET Core and React are established, widely-supported technologies. Azure SQL Database is a Microsoft managed service within the existing GDMS tenant. No novel or unvetted dependencies.
Team maintainability	Patching, backups, high availability, and scaling are fully managed by Microsoft. No DBA or DevOps specialist required. Dev team focuses on application code.
Browser compatibility	React SPA runs in any modern browser. No plugins required.

Strengths

- Lowest operational overhead of any database-hosting option — Microsoft manages backups, patching, HA, and scaling.
- Power BI connectivity is direct and well-documented; no gateway complexity.
- Azure AD integration is first-class with no additional configuration beyond app registration.
- .NET Core + React is a widely-adopted, well-documented stack with strong long-term support.
- Consistent with the team's existing skills and the GDMS toolchain.

Risks / Mitigations

Risk	Mitigation
Azure SQL Database cost at scale	Current workload (~140 users, few thousand records) is small. General Purpose tier is sufficient. Cost is predictable and low for this data volume.
React frontend requires build pipeline	Azure DevOps CI/CD pipeline handles this; no manual deployment process needed.

Candidate B — Not Recommended: TypeScript / NestJS + PostgreSQL + React

Overview

Layer	Technology
Frontend	React (TypeScript) — same as Candidate A
API	NestJS (TypeScript, Node.js runtime)
Database	PostgreSQL (self-managed or managed service, e.g. Azure Database for PostgreSQL)
Auth	Azure AD via passport-azure-ad or similar OIDC middleware
Hosting	Azure App Service or AKS on the GDMS GovCloud tenant
Source Control	Azure DevOps (GDMS GovCloud)

Evaluation Against Criteria

Criterion	Assessment
GovCloud boundary	Azure Database for PostgreSQL is available in Azure Government Cloud. Application hosting on App Service or AKS is within boundary.
Azure AD authentication	Supported via OIDC/OAuth2 middleware (e.g. passport-azure-ad), but requires more manual configuration than MSAL + Microsoft Identity Platform. Not first-class in the Node.js ecosystem the way it is in .NET.
Power BI connectivity	PostgreSQL is supported as a Power BI data source, but requires installing and configuring an on-premises data gateway in the GovCloud environment. Azure SQL connects natively without a gateway.
Concurrent user load (250)	Node.js handles concurrent I/O-bound workloads well. Acceptable for this scale.
GDMS-approved technologies	NestJS and PostgreSQL are open-source and widely used, but are not part of the Microsoft toolchain. Approval status within GDMS would need to be confirmed separately.
Team maintainability	Full TypeScript across frontend and backend is a coherence advantage, but the team's existing skills are in .NET — introducing NestJS adds a learning curve with no functional benefit for this application.
Browser compatibility	React SPA runs in any modern browser. No plugins required.

Weaknesses

- **Power BI gateway requirement:** PostgreSQL does not connect natively to Power BI in the cloud. An on-premises data gateway must be installed, configured, and maintained within the GovCloud environment — added infrastructure complexity that Azure SQL avoids entirely.

- **Azure AD integration overhead:** Microsoft Identity Platform / MSAL has first-class .NET support with well-maintained official libraries. The Node.js/NestJS equivalent requires more configuration and relies on community-maintained or less mature adapters.
- **Team skill alignment:** The dev team's existing experience is in .NET. Adopting NestJS introduces ramp-up time on a new runtime, framework conventions, and ORM (e.g. TypeORM or Prisma) without meaningfully improving the end product for this use case.
- **Toolchain consistency:** GDMS infrastructure, Azure DevOps pipelines, and the broader GovCloud toolchain are more naturally aligned with the Microsoft/.NET ecosystem. A Node.js backend introduces a toolchain mismatch with no offsetting benefit.
- **GDMS approval uncertainty:** Open-source non-Microsoft technologies may require additional approval steps within GDMS that .NET and Azure SQL do not.

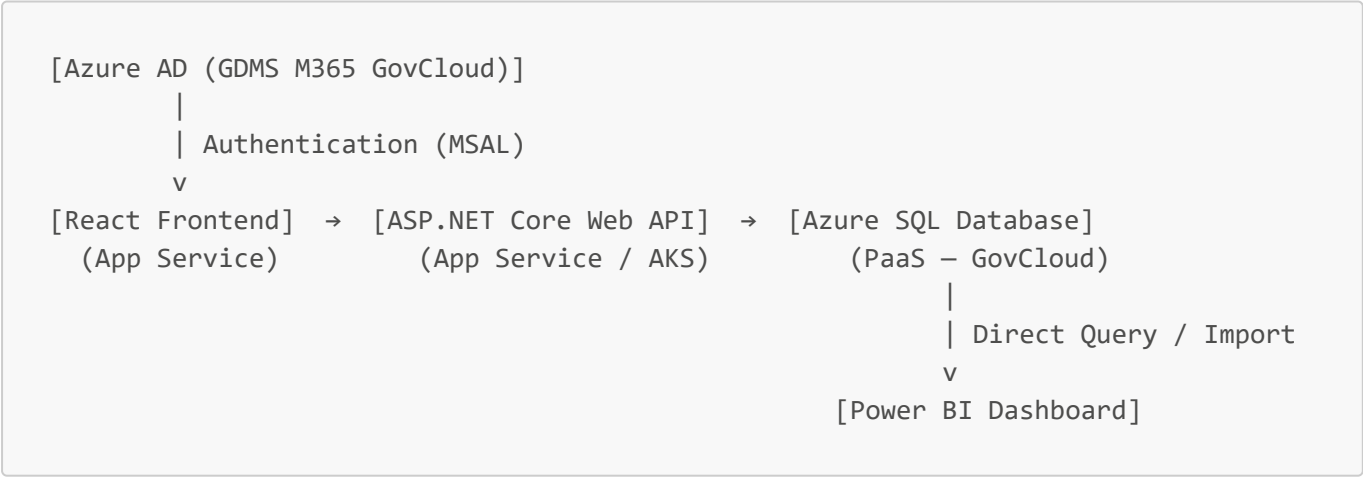
Recommendation

Candidate A — .NET Core + Azure SQL Database + React is recommended.

It satisfies all evaluated requirements, carries the lowest operational overhead for a small team, and supports Power BI connectivity natively without a gateway. Compared to Candidate B, it offers first-class Azure AD integration, better alignment with the team's existing .NET skills, and a more consistent fit with the GDMS GovCloud toolchain.

The database is hosted as an **Azure SQL Database managed service** (not SQL Server in a container), which eliminates the need for container/DBA management while still exposing standard SQL endpoints for Power BI connectivity.

Hosting Architecture Summary



All components hosted within the GDMS M365 GovCloud tenant. No data leaves the boundary.

Approval

Role	Name	Signature / Date
Author	Davis Blanch	2026-06-16

Role	Name	Signature / Date
Author	Ryan Clark	
Product Owner	Richard Patch	Pending

Per the requirements business rule: tech stack selection must be documented and approved by the product owner before build begins (NFR-5, AISAVING-130).